LAYERED PERMUTATION COPULAS ARE CLOSER TO THE UNIFORM LAW THAN THE DIAGONAL

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Abstract. Schrott conjectured that among all copulas on the unit square, the diagonal and the antidiagonal couplings are the farthest from the uniform law in the quadratic Wasserstein distance, at squared distance 1/10. By density of permutation copulas the conjecture is implied by a statement about n equally weighted grid points: for every permutation σ of $\{0,\dots,n-1\}$, the points $\big((i+\frac{1}{2})/n,(\sigma(i)+\frac{1}{2})/n\big)$ are no farther from the uniform law than the diagonal points. We prove this, with strict inequality, for every layered permutation, that is, every permutation obtained by cutting $0,\dots,n-1$ into runs of consecutive values and reversing some of the runs, other than the identity and the full reversal. The proof re-cuts the optimal partition of the identity on the union of the moved strips, writes the gain as a weighted sum of two-cell comparisons, and settles each comparison by a dispersive-order inequality between the laws of x_1+x_2 and x_1-x_2 on a diagonal band.

1. The problem

Let λ be Lebesgue measure on $Q=[0,1]^2$. A copula is a probability measure on Q with uniform marginals. Schrott [1] proved that among all couplings of n standard Gaussians, the ones farthest from the standard Gaussian on $\mathbb{R}^n$ in W_2 are the signed diagonals, and stated the uniform analogue as a conjecture.

Conjecture 1.1 ([1, Conjecture 1.1])**.** *For every copula π on Q, $W_2^2(\pi,\lambda)\le 1/10$, with equality for the diagonal coupling.*

For a probability measure μ on Q write $V(\mu)=\max\mathbb{E}[X\cdot Z]$ over couplings of $X\sim\lambda$ and $Z\sim\mu$. Since $W_2^2(\mu,\lambda)=\mathbb{E}|X|^2+\mathbb{E}|Z|^2-2V(\mu)$ and $\mathbb{E}|Z|^2=2/3$ for every copula, Conjecture 1.1 says that $V(\pi)\ge V(\text{diagonal})$ for every copula π.

Fix $n\ge 2$, put $a_i=(i+\frac{1}{2})/n$ for $0\le i\le n-1$, and for a permutation σ of $\{0,\dots,n-1\}$ let $\mu_\sigma=\frac{1}{n}\sum_i\delta_{p_i}$ with $p_i=(a_i,a_{\sigma(i)})$, and $A_n(\sigma)=V(\mu_\sigma)$. Both marginals of μ_σ are the uniform law on the grid $\{a_i\}$, so $\mathbb{E}|Z|^2$ does not depend on σ, and $A_n(\sigma)\ge A_n(\mathrm{id})$ says that μ_σ is no farther from λ than the diagonal points. Consider

$$(\mathrm{A}_n)\qquad A_n(\sigma)\ge A_n(\mathrm{id})\qquad\text{for every permutation }\sigma\text{ of }\{0,\dots,n-1\}.$$

Every copula π is a weak limit of measures μ_{σ_n}. Indeed, let $Z^1,\dots,Z^n$ be independent with law π, almost surely with distinct coordinates, and let σ_n send the rank of Z_1^j to the rank of Z_2^j (ranks counted from 0). Replacing each coordinate by $(\mathrm{rank}+\frac{1}{2})/n$ moves it by at most $\sup_t|F_n(t)-t|+\frac{1}{2n}$, where F_n is the empirical distribution function of that coordinate, which tends to 0 almost surely by the Glivenko–Cantelli theorem; and the empirical measure of the Z^j tends weakly to π. On Q weak convergence is W_2 convergence, and $|V(\mu)-V(\mu')|\le(\mathbb{E}|X|^2)^{1/2}W_2(\mu,\mu')$. Hence, if (A_n) holds for all n, then $V(\pi)=\lim_n A_n(\sigma_n)\ge\lim_n A_n(\mathrm{id})=V(\text{diagonal})=37/60$ for every copula π, the last limit by the same argument applied to the diagonal. So (A_n) for all n implies Conjecture 1.1.

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The map A_n is invariant under the eight symmetries of the square acting on permutation matrices, so $A_n(\mathrm{rev}) = A_n(\mathrm{id})$ for the full reversal $\mathrm{rev}(i) = n-1-i$. Numerically (A_n) holds for every permutation when $n \le 8$, with equality only at id and rev (Section 4).

Definition 1.2. A permutation σ is *layered* if there are integers $0 = k_0 < k_1 < \cdots < k_m = n$ such that, for each ℓ, σ maps $\{k_\ell, \dots, k_{\ell+1} - 1\}$ onto itself either identically or by reversal.

Theorem 1.3. *Let $n \ge 2$ and let σ be a layered permutation of $\{0, \dots, n-1\}$ other than* id *and* rev. *Then $A_n(\sigma) > A_n(\mathrm{id})$. The same holds for the images of σ under the symmetries of the square.*

For $n = 2$ every permutation is id or rev, so the theorem is about $n \ge 3$.

2. The identity and the re-cut

Write $s = x_1 + x_2$ and $v = x_1 - x_2$ for $x \in Q$. Under λ, $S = X_1 + X_2$ has density $L(s) = \min(s, 2-s)$ on $[0,2]$, and given $S = s$ the variable $V = X_1 - X_2$ is uniform on $[-L(s), L(s)]$. Let $0 = q_0 < q_1 < \cdots < q_n = 2$ be the quantiles of S at levels j/n and $T_j = \{x \in Q : q_j < s < q_{j+1}\}$.

Lemma 2.1. $A_n(\mathrm{id}) = \sum_j a_j \int_{T_j} s\,dx$.

Proof. A coupling of λ with μ_σ gives each point p_i a sub-probability $\gamma_i \le \lambda$ of mass $1/n$, and $\mathbb{E}[X \cdot Z] = \sum_i \int x \cdot p_i \, d\gamma_i$. For $\sigma = \mathrm{id}$, $x \cdot p_j = a_j s$, and with $a_0 < \cdots < a_{n-1}$ the sum $\sum_j a_j \int s\,d\gamma_j$ under $\sum_j \gamma_j = \lambda$, $\gamma_j(Q) = 1/n$, is largest when γ_j is λ on T_j (the rearrangement inequality applied to the law of S). □

Any partition of Q into regions $R_0, \dots, R_{n-1}$ of area $1/n$ gives the coupling $\gamma_i = \lambda|_{R_i}$, so $A_n(\sigma) \ge \sum_i \int_{R_i} x \cdot p_i \, dx$.

Consider a block of $r \ge 2$ consecutive values $k, \dots, k+r-1$ with $k + r \le n$, and let σ reverse it and fix every other value. Its moved points are $p_{k+i} = (a_{k+i}, a_{k+r-1-i})$, $0 \le i \le r-1$; they share $p_1 + p_2 = 2\bar{a}$ with $\bar{a} = a_k + (r-1)/(2n)$, and $p_1 - p_2 = w_i := (2i+1-r)/n$, increasing in i. Keep T_j for every fixed point, and cut $U = T_k \cup \cdots \cup T_{k+r-1} = \{q_k < s < q_{k+r}\}$ into the regions $R_0, \dots, R_{r-1}$ of area $1/n$ cut out by consecutive quantiles of v on U, from the smallest values of v up; give R_i to p_{k+i}. Since $x \cdot p = \frac{1}{2}\big(s(p_1+p_2) + v(p_1-p_2)\big)$, Lemma 2.1 gives $A_n(\sigma) - A_n(\mathrm{id}) \ge D$ with

$$D = \sum_{i=0}^{r-1} \int_{R_i} \Big(\bar{a}s + \tfrac{1}{2} w_i v\Big) dx - \sum_{j=k}^{k+r-1} a_j \int_{T_j} s\,dx = \frac{1}{n^2}\Big(\sum_{i=0}^{r-1} (i - \tfrac{r-1}{2})\,\nu_i - \sum_{i=0}^{r-1} (i - \tfrac{r-1}{2})\,m_i\Big),$$

where $m_i = n \int_{T_{k+i}} s\,dx$ is the mean of S on the strip T_{k+i} and $\nu_i = n \int_{R_i} v\,dx$ the mean of V on R_i.

Lemma 2.2. *For real numbers $c_0, \dots, c_{r-1}$, $\sum_i (i - \frac{r-1}{2}) c_i = \sum_{t=1}^{r-1} \frac{t(r-t)}{r}\,\Delta_t(c)$ with $\Delta_t(c) = \frac{1}{r-t}\sum_{i \ge t} c_i - \frac{1}{t}\sum_{i<t} c_i$.*

Proof. With $\bar{c}$ the mean, $\sum_i i(c_i - \bar{c}) = \sum_{t \ge 1} \sum_{i \ge t} (c_i - \bar{c})$, and $\sum_{i \ge t}(c_i - \bar{c}) = (r-t)\big(\frac{1}{r-t}\sum_{i \ge t} c_i - \bar{c}\big) = \frac{t(r-t)}{r}\Delta_t(c)$. □

So $n^2 D = \sum_{t=1}^{r-1} \frac{t(r-t)}{r}\big(\Delta_t(\nu) - \Delta_t(m)\big)$. Both ν and m are the means of a variable over its own quantile cells of mass $1/n$ on U: $\Delta_t(\nu)$ is the gap between the mean of V above and below its quantile at mass t/n on U, and $\Delta_t(m)$ the same gap for S.

3. A dispersive comparison on diagonal bands

Fix $0 \le \alpha < \beta \le 2$ with $(\alpha,\beta) \ne (0,2)$ and let $U = \{\alpha < s < \beta\}$, of area ρ. Let Φ_S and Φ_V be the distribution functions of S and V under $\lambda|_U$ (total mass ρ), Q_S, Q_V their inverses on $(0,\rho)$, and $\phi_S = L \circ Q_S$, $\phi_V = g \circ Q_V$ the densities read at mass level μ, where $g(y) = \frac{1}{2}|(\alpha,\beta) \cap (|y|, 2-|y|)|$ is the density of V under $\lambda|_U$. Write $\mu' = \rho - \mu$.

Lemma 3.1. *For $0 < \mu < \rho$, $\phi_V(\mu) \le \sqrt{2\min(\mu,\mu')} \le \phi_S(\mu)$. If $\alpha > 0$ there is $c > 0$ with $\phi_V(\mu) < \phi_S(\mu)$ for $0 < \mu < c$, and if $\beta < 2$ there is $c > 0$ with $\phi_V(\mu) < \phi_S(\mu)$ for $\rho - c < \mu < \rho$.*

Proof. Let $s = Q_S(\mu)$. If $s \le 1$, then $\mu = \int_\alpha^s L \le \int_0^s t\,dt = s^2/2$, so $\phi_S(\mu) = s \ge \sqrt{2\mu}$, strictly if $\alpha > 0$. If $s \ge 1$, then $\mu' = \int_s^\beta L \le (2-s)^2/2$ and $\phi_S(\mu) = 2 - s \ge \sqrt{2\mu'}$, strictly if $\beta < 2$. In both cases $\phi_S(\mu) \ge \sqrt{2\min(\mu,\mu')}$.

For the left inequality, g is even. For $y \in [-1,0]$ the interval $(-y, 2+y)$ grows with y and each of its ends moves at unit speed, so g is nondecreasing on $[-1,0]$, $g(-1) = 0$, and $g(y) - g(z) \le y - z$ for $-1 \le z \le y \le 0$. Hence for $y = Q_V(\mu) \le 0$,

$$\mu = \int_{-1}^{y} g(z)\,dz \ge \int_{-1}^{y} \big(g(y) - (y-z)\big)_+ dz \ge \tfrac{1}{2} g(y)^2,$$

so $\phi_V(\mu) \le \sqrt{2\mu}$, and $\mu \le \mu'$ by the symmetry of g. The case $y \ge 0$ is the mirror image.

For the strict inequality let $\alpha > 0$. If $\alpha < 1$, put $c = \min(\Phi_S(1), \rho) > 0$: for $\mu < c$ we have $Q_S(\mu) < 1$ and $\phi_S(\mu) > \sqrt{2\mu} \ge \phi_V(\mu)$ by the first paragraph. If $\alpha \ge 1$, put $c = \rho/2$: then $Q_S(\mu) \ge 1$, and for $\mu < c$ we have $\mu < \mu'$, so $\phi_S(\mu) \ge \sqrt{2\mu'} > \sqrt{2\mu} \ge \phi_V(\mu)$. The case $\beta < 2$ follows by applying the map $(x_1, x_2) \mapsto (1 - x_2, 1 - x_1)$, which preserves λ, fixes v, sends s to $2 - s$ and the band (α,β) to $(2-\beta, 2-\alpha)$, and reverses the mass levels of S. □

Proposition 3.2. *Let $n \ge 3$, $2 \le r \le n-1$, $0 \le k \le n - r$. Then $D > 0$ for the reversal of the block $k, \dots, k + r - 1$.*

Proof. Take $\alpha = q_k$, $\beta = q_{k+r}$; since $r < n$, $(\alpha,\beta) \ne (0,2)$. Since $Q_S' = 1/\phi_S$ and $Q_V' = 1/\phi_V$, Lemma 3.1 shows that $h = Q_V - Q_S$ is nondecreasing on $(0,\rho)$. For $\tau = t/n$,

$$\Delta_t(\nu) - \Delta_t(m) = \frac{1}{\tau(\rho - \tau)} \int_0^\tau \int_\tau^\rho \big(h(\mu) - h(\mu_0)\big)\,d\mu\,d\mu_0 \ge 0.$$

If $\alpha > 0$, Lemma 3.1 gives $c > 0$ with h strictly increasing on $(0,c)$, so $h(\mu) > h(\mu_0)$ whenever $\mu_0 < \min(\tau, c)/2$ and $\mu > \tau$; the integral is then positive for every t. If $\alpha = 0$ then $\beta < 2$, h is strictly increasing on some $(\rho - c, \rho)$, and $h(\mu) > h(\mu_0)$ whenever $\mu_0 < \tau$ and $\mu > \rho - \min(\rho - \tau, c)/2$. By Lemma 2.2, $n^2 D > 0$. □

Proof of Theorem 1.3. Let the reversed runs of σ be $B_1, \dots, B_m$, of sizes $r_\ell \ge 2$. If some $r_\ell = n$ then $\sigma = \mathrm{rev}$; otherwise each $r_\ell \le n - 1$, and $m \ge 1$ since $\sigma \ne \mathrm{id}$. The unions U_ℓ of the strips of the runs are disjoint, so performing the re-cut of Section 2 inside every U_ℓ and keeping the other strips is a partition of Q into regions of area $1/n$ matched to the points of μ_σ. Its value is $A_n(\mathrm{id}) + \sum_\ell D_\ell$, and each $D_\ell > 0$ by Proposition 3.2. The symmetries of the square preserve λ and the grid, so A_n is invariant under them. □

Remark 3.3. For $r = 2$ at the corner ($k = 0$ or $k = n - 2$, $n \ge 4$) the bound is exactly $n^2 D = \frac{2\sqrt{2}-2}{3}\, n^{-1/2} \approx 0.276\, n^{-1/2}$, while the measured gap $n^2(A_n(\sigma) - A_n(\mathrm{id}))$ is about $0.344\, n^{-1/2}$ ($n = 8, 12, 16$). The corner transposition is the cheapest move, so the margin in (A_n) near the identity is of order $n^{-5/2}$.

4. Computations

All computations use exact moments of S and one-dimensional quadrature of the law of V on a band.

(1) Check of (A_n) over every permutation for $n \le 8$ (one per class under the symmetries of the square; 5,282 classes at $n = 8$), with A_n computed by a smoothed semi-discrete dual on a 40×40 grid of cells. No permutation falls below the identity; the smallest positive gap is the corner adjacent transposition ($n^2(A_n(\sigma) - A_n(\mathrm{id})) = 0.1215$ at $n = 8$), and the first non-layered permutations are corner 3-cycles (0.3944 at $n = 8$).
(2) The bound D for every block reversal, $3 \le n \le 48$, $2 \le r \le n - 1$, every k: all positive, with minimum at the corner adjacent transposition, $n^{5/2}D = 0.2761$, matching the closed form above. Quadrature error below 10^{-8}.
(3) The identity of Lemma 2.2 applied to D, checked against the direct evaluation for $n = 5, 8, 13$ and every (k, r): difference below 1.1×10^{-8}; every term $\Delta_t(\nu) - \Delta_t(m)$ is positive.

5. What remains

Theorem 1.3 covers the layered permutations and their images under the symmetries of the square. For a non-layered σ the moved points do not share a value of $p_1 + p_2$, and a cut of a band along one direction is not enough on the whole square: cutting Q into quantile cells of $s\cos\theta + v\sin\theta$ and matching the cells optimally never beats the strips for any θ when $n = 4$. Inside a window of a few strips the same directional cut does give positive bounds for every indecomposable block permutation that was tested ($n = 8$, windows of 3, 4, 5 strips), so a proof of (A_n) along these lines would need cuts whose direction varies across the square.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the reduction, the re-cut construction and the proofs, wrote and ran the computations of Section 4, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the computations of Section 4, with its output, is available at `https://github.com/yspennstate/wasserstein-layered-permutation-copulas`.

References

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